Label-Free, Dependency-Aware Channel Selection for Coupled Multi-Channel Sensor Data

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***Abstract*—A method is presented for selecting a small, informative subset of a multi-channel sensor's original channels without using labels. A group-structured autoencoder is trained to reconstruct all channels; each latent factor is then perturbed, and the change in every channel's reconstruction gives a label-free estimate of its relevance. A relevance–redundancy rule turns these estimates into an ordered shortlist of real channels. The same procedure was applied to two very different modalities, changing only the encoder. On a wearable-sensor activity benchmark (three inertial units, 27 channels, leave-one-subject-out), retaining ten channels preserved macro-F1 to within 0.04 of the full set, equalled a supervised mutual-information selector, and was far more stable across subjects than variance ranking. On biomedical imaging the same method removed 58–95% of channels while accuracy was maintained or improved (88.2% to 95.2% on lichen tissue; 79.8% to 85.6% on collagen sponges). A single label-free principle thus delivers strong, interpretable channel reduction across very different sensors.**

***Index Terms*—unsupervised feature selection, channel selection, conditional relevance, autoencoders, interpretability, human activity recognition.**

I. Introduction

Complex sensors can have data coming from many channels that overlap heavily and arrive in natural groups. For cost, power, transmission bandwidth, and interpretability, only a handful of the actual channels are often required, rather than a learned mixture of them. The central difficulty is to determine which channels to keep when no labels are available.

Existing selectors present a trade-off. Unsupervised filters such as variance ranking or principal component analysis (PCA) score each channel by its own marginal statistics and disregard how channels depend on one another, so several channels carrying the same information tend to be retained together. Dependency-aware selectors such as minimum-redundancy maximum-relevance (mRMR) and joint mutual information (JMI) account for this redundancy but require labels and density estimation. The region of the design space that is simultaneously label-free, dependency-aware, and discrete—returning real channels rather than a projection—remains sparsely populated. This paper addresses that region with a method built on a group-structured autoencoder and perturbation-based attribution, and it is shown that the identical method operates across two very different sensing modalities by changing only its front-end encoder. The contributions are: (i) a label-free, dependency-aware, discrete channel selector; (ii) a modality-agnostic selection engine that adapts to a new domain through the encoder alone; and (iii) verification on two modalities, matching a supervised selector on wearable-sensor data without labels and removing the majority of channels on biomedical imaging without loss of accuracy.

TABLE I
Positioning among representative selector families.

Family	Labels needed	Models dependency	Real channels
Variance / PCA	no	no	yes/no
Laplacian, MCFS	no	weak	yes
mRMR, JMI	yes	yes	yes
Concrete AE	no	implicit	yes
This method	**no**	**yes**	**yes**

II. Related Work

Channel and feature selection methods fall into filter, wrapper, and embedded families. Variance ranking and the Laplacian score [4] are unsupervised but marginal: they ignore inter-channel dependency and tend to retain redundant channels. Dependency-aware selectors model conditional relevance—a channel's usefulness given those already retained—through criteria such as mRMR and JMI [2], [3], but these are supervised. Among unsupervised deep methods, the concrete autoencoder [1] learns a differentiable discrete selection layer, and in hyperspectral imaging attention networks such as BS-Net [5] rank spectral bands. The present method is closest to this lineage; its distinguishing combination is perturbation-based reconstruction sensitivity as the relevance signal on a group-structured representation, an explicit relevance–redundancy step, and demonstrated transfer across modalities (Table I).

III. Method

A. Problem Formulation

A multi-channel observation is organised into G groups. Group g contributes C_g channels measured over a regular axis (time for sensor streams, space for images) of length T, so one windowed sample is $X_g \in \mathbb{R}^{T\times C_g}$ and the full sample is $X = (X_1, \ldots, X_G)$. The objective is to return an ordered subset $S \subseteq \{(g,c)\}$ of the original channels, with $|S| = K$, that is informative and non-redundant, using no labels.

B. Group-Structured Autoencoder

Each group is encoded independently and the per-group features are fused by averaging into a shared latent representation, then decoded per group:

$$z = \frac{1}{G}\sum_{g=1}^{G} E_g(X_g), \qquad \hat{X}_g = D_g(z), \tag{1}$$

where E_g, D_g are the encoder and decoder for group g and $z \in \mathbb{R}^{d\times T}$. Averaging ties the groups into one joint code and keeps the latent size independent of G. Training minimises reconstruction error only, $\mathcal{L} = \sum_g \|X_g - D_g(z)\|_2^2$, with no labels. The encoder is the only modality-specific component: temporal convolutions for sensor streams, spatial convolutions for images.

C. Perturbation-Based Relevance

The latent coordinates are ranked by their variance across samples and the top m are retained. For a retained coordinate j and magnitude δ, the latent is shifted, $z^{(j,\delta)} = z + \delta e_j$, and the relevance of channel c in group g is the accumulated change in its reconstruction:

$$I_{g,c} = \sum_{j,\delta} w_j \, \frac{1}{NT}\sum_{n,t}\Big|\big[D_g(z^{(j,\delta)}) - D_g(z)\big]_{n,t,c}\Big|, \tag{2}$$

averaged over samples and axis positions, with w_j weighting each coordinate by its importance. A channel whose reconstruction is sensitive to the latent factors is one the joint representation depends upon.

D. Normalisation and Relevance–Redundancy Selection

Relevances are normalised within each group, $\tilde{I}_{g,c} = I_{g,c}/\max_{c'} I_{g,c'}$, which is preferable to dividing by the data variance when the discriminative channels themselves have high variance. Channels are then chosen by a maximal-marginal-relevance rule:

$$c^\star = \arg\max_{(g,c)\notin S}\Big[\tilde{I}_{g,c} - \lambda \max_{s\in S} \operatorname{sim}\big((g,c), s\big)\Big], \tag{3}$$

where sim is the cosine similarity between channel data profiles and λ controls the redundancy penalty. Selection begins with the most relevant channel and proceeds greedily, giving an ordered list truncated to any budget K (Fig. 1).

E. Cross-Domain Instantiation

The selection engine operates only on groups, channels, and the latent, so the same engine is instantiated in two domains by substituting the encoder (Table II). For wearable sensors a group is a body-worn inertial unit, a channel is one sensor axis, the regular axis is time, and a one-dimensional convolutional encoder is used. For fluorescence imaging a group is an excitation wavelength, a channel is an emission band, the regular axis is the two-dimensional image plane, and a spatial convolutional encoder is used.

TABLE II
THE SAME METHOD IN TWO DOMAINS; ONLY THE ENCODER CHANGES.

Concept	Wearable sensors	Biomedical imaging
group	body-worn IMU	excitation wavelength
channel	IMU axis (acc/gyro/mag)	emission band
axis	1D time	2D space
encoder	Conv1D	Conv2D / Conv3D
engine	identical	identical

IV. Experimental Setup

The wearable-sensor dataset was PAMAP2: three inertial measurement units (IMUs) on the hand, chest, and ankle, each contributing nine channels (three-axis accelerometer, gyroscope, and magnetometer), for 27 channels in three groups, with 38,856 one-second windows over eight subjects and twelve activities. A strict leave-one-subject-out (LOSO) protocol was used: the autoencoder was trained without labels on the remaining subjects, the engine produced an ordered channel list, and the subset was verified by a k-nearest-neighbour classifier (k=5); macro-F1 is reported as the LOSO mean. The proposed method was compared against a supervised mutual-information selector, a variance baseline, and random selection. The biomedical modality consisted of multi-excitation fluorescence image cubes: a lichen-tissue dataset (192 bands, four classes) and a collagen-sponges dataset (158 bands, three classes), processed by the identical engine with a spatial encoder and verified by per-pixel classification accuracy.

V. Results and Discussion

A. Wearable-Sensor Activity Recognition

The relationship between the number of retained channels and recognition performance is shown in Fig. 2. Reducing from 27 to 10 channels (a 63% reduction) preserved macro-F1 to within 0.04 of the full-set value (0.68 versus a 0.72 ceiling); at the same budget the proposed method equalled the supervised mutual-information selector (0.68) and exceeded variance (0.59). Across budgets, the proposed method tracked the supervised selector and remained above the variance baseline. The selected channels were also more stable across subjects than variance selection (Fig. 3), which matters when a fixed sensor set must be chosen before deployment. For full transparency, on this benchmark no selector, including the supervised one, surpassed random selection, as the 27 channels are highly redundant; the informative comparisons are therefore the agreement with a supervised selector obtained without labels and the improved stability over variance.

B. Biomedical Imaging

On the lichen-tissue dataset, 89.4% classification accuracy was reached using only nine of 192 bands (a 95% reduction), exceeding the 88.2% obtained with all bands; 95.2% was reached with 80 bands. On the collagen-sponges dataset, 30 of 158 bands raised accuracy from 79.8% to 85.6% (Fig. 4).

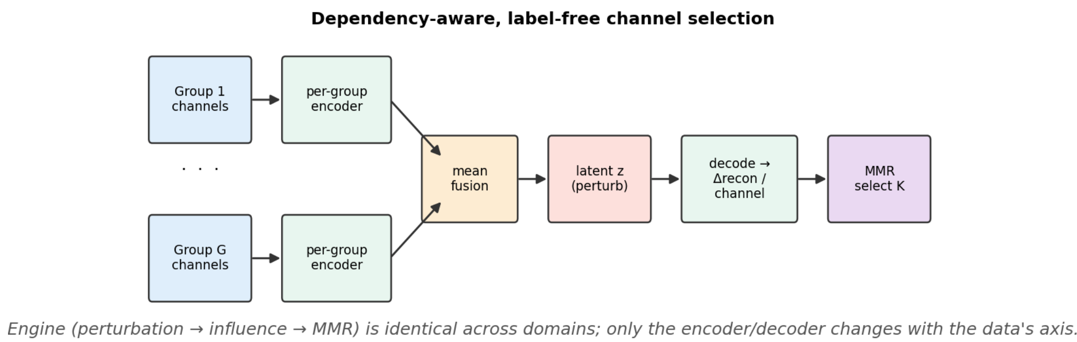

Fig. 1. The three-stage pipeline. A group-structured autoencoder is trained by reconstruction; latent perturbations yield a per-channel relevance; and a relevance–redundancy rule selects an ordered subset. The selection engine is identical across domains; only the encoder and decoder change to match the data's regular axis.

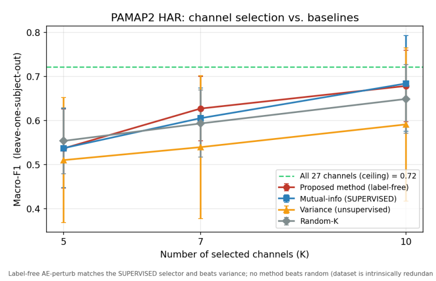

Fig. 2. Macro-F1 versus number of retained channels (LOSO). The proposed method (red) tracks the supervised selector (blue) and stays above variance (orange); the dashed line is the full-set ceiling.

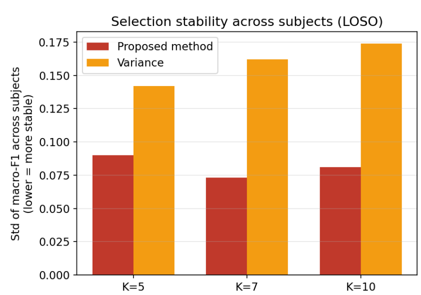

Fig. 3. Stability of the selected set across subjects (lower is better). The proposed method is markedly more consistent than variance selection.

In this high-redundancy regime the same principle yields large reductions with no loss, and often a gain, in accuracy.

Together these results show that the value of a selector depends on the redundancy of the data, and that modelling redundancy explicitly is what separates the proposed method from variance ranking and explains its improved stability.

VI. Conclusion

A single unsupervised, dependency-aware selection principle delivered strong, interpretable channel reduction across two very different modalities, requiring only a change of encoder: it matched a supervised selector on wearable sensors without labels and selected more stably than variance ranking, and

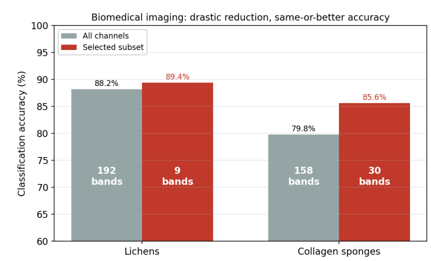

Fig. 4. Biomedical imaging: drastic channel reduction maintains or improves accuracy. Band counts are shown on the columns.

it removed 58–95% of channels on biomedical imaging while accuracy was maintained or improved. Future work will target a higher-redundancy activity benchmark and a comparison with differentiable unsupervised selectors.

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References

[1] M. F. Balın, A. Abid, and J. Zou, "Concrete autoencoders: Differentiable feature selection and reconstruction," in *Proc. ICML*, 2019.
[2] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, 2005.
[3] G. Brown, A. Pocock, M.-J. Zhao, and M. Luján, "Conditional likelihood maximisation: A unifying framework for information theoretic feature selection," *J. Mach. Learn. Res.*, vol. 13, pp. 27–66, 2012.
[4] X. He, D. Cai, and P. Niyogi, "Laplacian score for feature selection," in *Proc. NeurIPS*, 2005.
[5] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, 2020.
[6] A. Reiss and D. Stricker, "Introducing a new benchmarked dataset for activity monitoring," in *Proc. IEEE Int. Symp. Wearable Computers (ISWC)*, 2012, pp. 108–109.